

1. Abstract

This thesis studies the mechanisms involved in perceptual decision tasks by combining behavioral analysis, computational modeling, machine learning techniques and electroencephalogram (EEG) analysis. The project is based on a perceptual task in which participants repeatedly classify sequences of visual stimuli, allowing the study of how previous decisions influence the integration of new sensory evidence.

The main objective is to understand perceptual confirmation bias, understood as the tendency to prioritize information consistent with past decisions while reducing the weight of contradictory visual information. To study this, mixed models and evidence accumulation models are applied to behavioral data. Later, machine learning approaches and EEG analysis will also be included to study the predictive value of behavioral and neural variables.

The results obtained so far show a strong tendency to repeat prior decisions across successive stimulus repetitions. This effect seems to depend on subjective confidence and changes in sensory evidence integration rather than simple response repetition strategies. In addition, the computational models suggest that earlier choices alter the accumulation of evidence in a way that is consistent with perceptual confirmation bias mechanisms.

Overall, this work contributes to understanding the cognitive and neural processes underlying perceptual decision-making and in the biases related to perceptual information integration.

Keywords - computational neuroscience, perceptual decision-making, confirmation bias, EEG, machine learning, evidence accumulation, drift diffusion model, metacognition, confidence, changes of mind.

2. Introduction and state of the art

2.1 Perceptual decision-making and evidence accumulation

Perceptual decisions are one of the main research areas in cognitive neuroscience and experimental psychology because it helps explain how the brain transforms ambiguous sensory information into clear decisions. In these tasks, individuals accumulate sensory evidence over time until they choose between different possible alternatives. This process has been widely described using computational evidence accumulation models, especially the Drift Diffusion Model (DDM) (Ratcliff, 1978; Ratcliff & McKoon, 2008).

The DDM describes decision-making as a dynamic process in which evidence is gradually accumulated until a decision threshold is reached. These models have shown a strong ability to explain both participants' decisions and their response times, making them an important tool for studying human cognition.

Traditionally, many experimental paradigms assumed that perceptual decisions were mostly independent across trials. However, recent studies have shown that previous decisions can influence the interpretation of new sensory evidence, even when the stimuli are independent. These effects, known as choice history biases or serial dependencies, reflect a tendency of the perceptual system to incorporate information from prior choices when processing new perceptual stimuli.

2.2 Choice history bias and confirmation bias

Several studies have shown that perceptual decisions do not depend only on current sensory evidence, but also on recent response history and earlier choices. This phenomenon, known as choice history bias, describes the tendency of individuals to repeat previous decisions even when the stimuli are independent from each other (Urai et al., 2019). These findings suggest that previous experiences shape how people interpret new sensory information.

Recent research suggests that these biases are not simply caused by a general tendency to repeat responses, but may appear directly during the process of sensory evidence integration. In particular, different studies have shown that information consistent with past responses tends to receive more weight during evidence accumulation, while contradictory information is processed less strongly (Talluri et al., 2018; Rollwage et al., 2020).

Different computational models have tried to explain the origin of these biases. Some authors propose mechanisms based on self-consistent bayesian inference (Luu & Stocker, 2018), while others suggest that the bias emerges directly during evidence accumulation through selective overweighting of information that is

consistent with previous responses (Talluri et al., 2018; Bronfman et al., 2015). In this context, sequential evidence accumulation models are useful for studying different possible bias effects involved in sensory decision-making.

This project is part of this research line and studies how prior decisions modulate the integration of sensory evidence when participants repeatedly process the same sequence of visual stimuli. This paradigm allows us to study how confirmation bias affects the stability and revision of previous perceptual decisions.

2.3 Metacognition, confidence and changes of mind

The ability to review previous decisions is an important aspect of perceptual decision-making. These processes, known as changes of mind, reflect the possibility that new sensory evidence can modify previously made decisions.

The probability of maintaining or changing a decision seems to depend strongly on the level of confidence associated with that response. Perceptual confidence acts as an internal signal about the subjective probability of having made a correct decision and can influence how new information is integrated after an initial choice (Peters et al., 2017; Rollwage et al., 2020). Different studies have shown that decisions made with high confidence tend to be more resistant to revision and promote a biased integration of evidence consistent with previous decisions. These processes are closely related to metacognition, understood as the ability of individuals to evaluate the accuracy of their own decisions (Fleming & Dolan, 2012). Recent studies suggest that metacognitive efficiency may partially contribute to confirmation biases by helping individuals detect errors and update decisions when facing contradictory evidence (Rollwage et al., 2020).

In this context, this study investigates how metacognitive factors such as confidence and metacognitive efficiency affect the stability and revision of repeated perceptual decisions.

2.4 Justification of the current project

This thesis studies how previous decisions shape the processing of sensory evidence when participants repeatedly process the same sequence of visual stimuli. Unlike many classical perceptual decision-making paradigms, where each trial is considered independent, this experimental design allows the study of how individuals maintain or revise previous perceptual interpretations when facing partially redundant information. In this task, participants observe sequences of oriented stimuli and repeatedly classify them as mainly cardinal or diagonal. The controlled repetition of these sequences makes it possible to investigate how strongly initial decisions modulate the later processing of new sensory evidence.

This paradigm is especially useful for studying perceptual confirmation bias. It helps distinguish between a simple tendency to repeat previous responses and a biased processing of sensory evidence consistent with the earlier decision.

To understand these processes, the project combines behavioral analyses, mixed models and computational evidence accumulation models. In later stages, machine learning approaches and EEG analysis will also be included to explore the predictive and neural processes involved in visual decision-making.

3. Objectives

3.1 Main objective

The main objective of this project is to investigate how previous decisions shape the integration of new sensory evidence during repeated perceptual tasks, as well as to better understand the mental processes and computational models that explain this phenomenon. In particular, the project aims to study whether the tendency to maintain earlier responses is caused by a confirmation bias, meaning a tendency to interpret new information in a way that supports what we previously decided.

In addition, the project aims to investigate how factors such as confidence in our decisions modulate whether we maintain or change them. It also wants to evaluate different computational and predictive models to describe and predict participants' behavior.

3.2 Specific objectives

1. To behaviorally analyze the tendency of participants to repeat or modify perceptual decisions across successive repetitions of the same visual stimulus.
2. To analyze how variables such as sensory evidence, confidence and previous decisions influence participants' behavior using mixed models.
3. To determine whether the observed biases can be explained by simple response repetition mechanisms or whether they reflect an asymmetric processing of sensory evidence consistent with confirmation bias processes.
4. To apply Hierarchical Sequential Sampling Models (HSSM) to compare different computational bias mechanisms, including starting point bias and drift-rate bias.
5. To evaluate, through simulations and parameter recovery analyses, the ability of computational models to reliably recover the parameters generating participants' behavior.
6. To implement machine learning techniques to study how much participants' decisions can be predicted from behavioral variables and parameters derived from computational models.
7. To analyze, in later stages of the project, the neural correlates associated with evidence integration and confirmation bias using EEG.
8. To explore the combined predictive value of behavioral and neural variables for anticipating participants' decisions.

4. Changes with respect to the initial plan

During the development of the project, some aspects of the original plan were refined in order to better adapt the work to the real characteristics of the data and to improve the interpretation of the results. These changes did not modify the main research question, but helped to study the mechanisms involved in perceptual confirmation bias more precisely.

In the first stage, the behavioral analysis was carried out using conventional models. Later, mixed models were incorporated to better represent individual variability. This change allowed more robust estimations that were more consistent with the structure of the experiment.

In addition, the progress of the project showed the need to complement the behavioral analysis with computational accumulation models. For this reason, HSSM were included to compare different computational bias mechanisms related to changes in the starting point and drift rate.

Finally, the development of the project also involved a reorganization of the analysis pipeline, giving priority to behavioral and computational analyses before moving to later stages based on machine learning and EEG analysis. At the same time, the theoretical framework related to confirmation bias, sensory information accumulation and metacognition was further developed, allowing a more coherent formulation of the project objectives.

5. Methodology

5.1 Experimental design

The project is based on a perceptual decision-making task designed to study how previous decisions affect how new evidence is processed. In this task, participants observed sequences of oriented visual stimuli (Gabor patches) and had to classify them as mainly cardinal or diagonal.

Each sequence consisted of six stimuli presented one after another. Task difficulty was manipulated using a continuous variable called the decision variable (DV) which represented how close the sequence was to cardinal or diagonal orientations. Values close to zero corresponded to ambiguous stimuli, while more extreme values represented perceptually clearer sequences.

The main feature of the paradigm was the controlled repetition of sequences in the same trial. After making an initial decision, participants observed the same sequence or a very similar sequence again and had to give a second response. This design made it possible to study how previous decisions influenced the later processing of new sensory evidence.

In addition to categorical decisions, the experiment also recorded response times and subjective confidence ratings associated with each decision. Task difficulty was dynamically adjusted using a 1-up/2-down adaptive staircase procedure, designed to maintain relatively stable accuracy levels across participants.

5.2 Dataset and preprocessing

The analysis carried out so far has focused on the behavioral data obtained during the perceptual task, including decisions, response times, subjective confidence ratings and variables derived from the experimental design. Before the statistical and computational analyses, a preprocessing procedure was applied to ensure data quality. Outlier response times were filtered using criteria defined individually for each participant based on the mean and standard deviation of their response times. After this filtering process, only a small proportion of trials was removed compared to the original dataset.

In addition, several derived variables were created for later analyses, including indicators of decision repetition or change (switch/repeat behavior), variables related to previous response history and metacognitive measures based on subjective confidence and meta-d' estimations.

5.3 Behavioral analysis

The behavioral analysis aimed to study how previous decisions, sensory evidence and metacognitive variables such as confidence, shape participants' perceptual behavior across the different stimulus repetitions.

In the first stage, descriptive analyses were carried out to identify general behavioral patterns in the participants. These analyses included response times, accuracy, probability of repeating decisions, psychometric curves and the relationship between subjective confidence and the tendency to maintain or change responses.

Later, linear mixed models were incorporated to analyze how variables such as sensory input, previous decisions, subjective confidence and recent response history affected different behavioral measures, including the probability of repeating decisions, changes of mind and perceptual accuracy.

This approach allowed a more precise analysis of the behavioral operations involved in perceptual confirmation bias and later complemented the study with computational accumulation models.

5.4 Drift Diffusion Modeling with HSSM

To investigate the hidden mechanisms behind the behavioral biases observed in the task, a computational approach based on HSSM was included. These models make it possible to describe both participants' decisions and their response times within the DDM framework.

In particular, models in which previous decisions modified the starting point of evidence accumulation were compared with models in which earlier choices directly altered the integration of new sensory stimuli (drift-rate bias).

Before applying the models to the real data, simulations and parameter recovery analyses were carried out to evaluate whether the models could reliably recover the parameters generating the behavior.

After that, the models were fitted to the real behavioral data. Model comparison was performed using fitting and predictive performance metrics such as WAIC (Widely Applicable Information Criterion) and LOO (Leave-One-Out Cross-Validation).

6. Results

6.1 Behavioral results

6.1.1 General performance and response times

As a first approach to participants' behavior, response times and overall accuracy were analyzed across the different experimental conditions. Response time did not vary significantly across repetitions or trial types. Overall accuracy also remained relatively stable across conditions and repetitions, indicating that the adaptive difficulty procedure worked properly.

6.1.2 Decision repetition and confirmation bias

The main goal of the behavioral analysis was to determine to what extent initial decisions impacted later perceptual responses. For this reason, the probability that participants repeated their previous decision across the different stimulus repetitions was analyzed.

The results showed a strong tendency to maintain past decisions during successive repetitions of the same sequence, reaching repetition probabilities close to 85% (Figure 1). This effect appeared consistently across participants and was highly significant.

To determine whether this stability could simply be explained by perceptual memory or stimulus recognition, the repeat and nonrepeat conditions were compared. The results showed very similar repetition levels in both conditions, suggesting that the tendency to maintain previous responses did not depend only on the exact repetition of the visual stimulus.

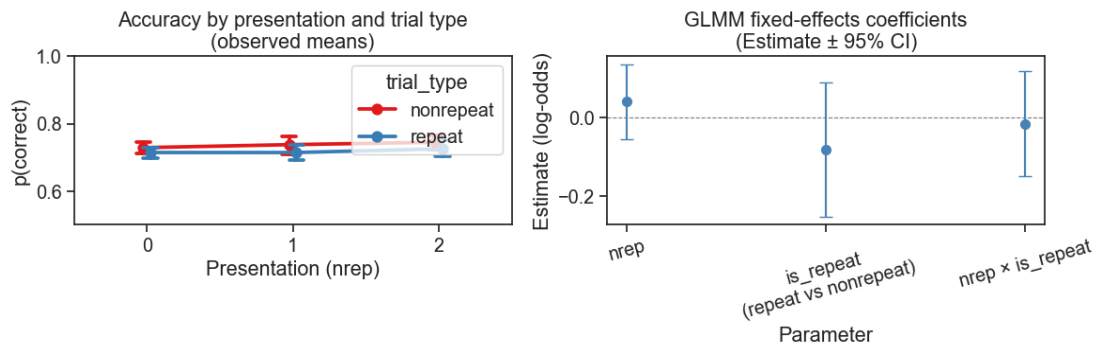


Figure 1. Accuracy across stimulus repetitions and trial types (repeat vs nonrepeat), together with GLMM fixed-effects coefficients.

6.1.3 Confidence, metacognition and changes of mind

To study the role of metacognitive processes in the stability and revision of perceptual decisions, the relationships between subjective confidence, changes of mind and decision repetition behavior were analyzed.

The results showed that confidence associated with the previous decision significantly contributed to the probability of maintaining or changing a later response (Figure 2). In general, participants tended to keep their decisions when previous confidence was high, while changes of mind were associated with considerably lower confidence levels.

In contrast, metacognitive efficiency was not strongly associated with the tendency to repeat previous decisions. However, trial-by-trial confidence strongly modulated confirmation bias, such that participants were more likely to repeat previous responses when confidence in the preceding decision was high.

In addition, changes of mind did not lead to higher perceptual accuracy, suggesting that revising a previous decision was not always beneficial under conditions of ambiguous or redundant sensory evidence.

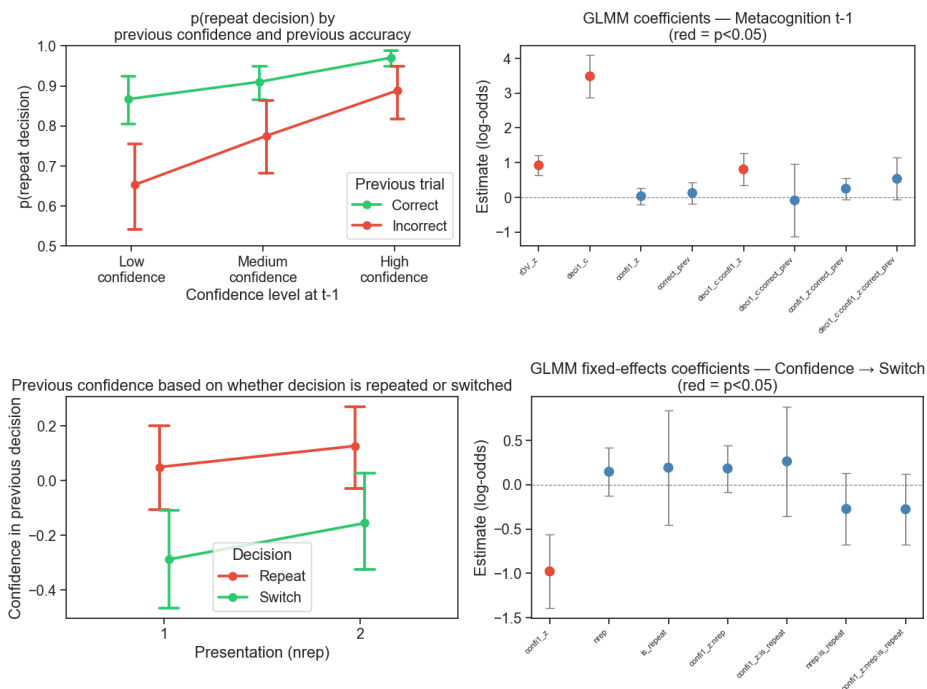


Figure 2. Relationship between confidence, metacognition and decision repetition behavior.

6.1.4 Asymmetric processing of sensory evidence

To evaluate whether the observed bias reflected only a general tendency to repeat responses or also affected the processing of sensory input, the contribution of the different stimuli in the sequence was analyzed while controlling for participants' previous decisions.

The results showed that the previous decision had a much stronger influence than any individual stimulus, indicating a strong effect of earlier responses on later decisions (Figure 3). In addition, after controlling for this repetition bias, stimuli that were incongruent with the previous decision tended to provide more residual information than congruent stimuli.

This pattern suggests that much of the congruent information was already captured by the previous decision itself, while contradictory information provided information that could lead participants to change their final decision. Overall, these findings support the idea that earlier decisions actively shape the later integration of sensory evidence, in a way that is consistent with perceptual confirmation bias mechanisms.

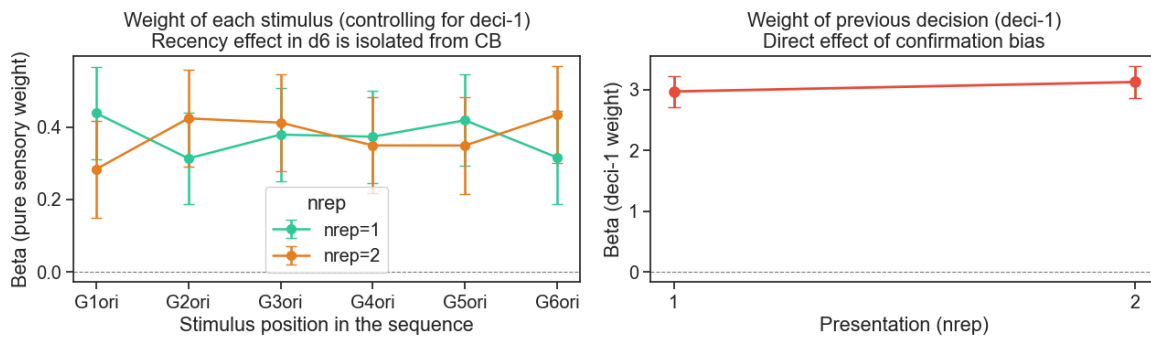


Figure 3. Contribution of sensory evidence and previous decisions to later perceptual responses.

6.1.5 Summary of behavioral results

Overall, the behavioral results showed a strong influence of previous responses on the integration of new sensory inputs during repeated perceptual decision-making tasks. Participants tended to maintain previous decisions across successive stimulus repetitions and an asymmetric processing of evidence that was congruent with the previous response.

These findings motivated the later application of computational accumulation models to investigate the hidden mechanisms underlying the observed biases.

6.2 Results HSSM

6.2.1 Simulations and parameter recovery

Before applying the HSSM models to the real dataset, simulations and parameter recovery analyses were performed to evaluate whether the models could reliably recover known parameters introduced during data generation. The results showed satisfactory recovery of the main model parameters, particularly those related to drift rate and starting point, providing a methodological validation of the inference procedures used.

6.2.2 Fitting models to real behavioral data

After validating the models through simulations and parameter recovery, the HSSM models were fitted to the real behavioral data obtained during the perceptual task.

The main goal of this analysis was to investigate which computational mechanisms could better explain the behavioral biases observed in participants.

On the one hand, models were tested where the previous decision modified the starting point of the decision process. On the other hand, models were evaluated in which previous responses altered the drift rate, suggesting that prior choices guide how new sensory evidence was accumulated during decision-making.

The models were fitted using hierarchical bayesian inference and included variables related to sensory evidence, initial decisions and subjective confidence.

HSSM on real CJ_EEG data
Red line = absence of bias

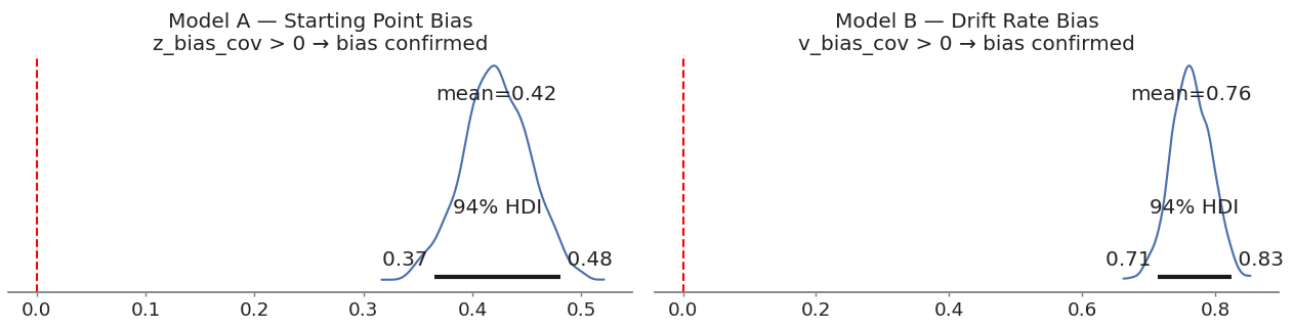


Figure 4. Comparison between HSSM models including starting point bias and drift-rate bias.

6.2.3 Model comparison and computational interpretation

To determine which computational mechanism better explained the observed behavioral patterns, a formal comparison between HSSM models was performed using bayesian model comparison metrics, including WAIC and LOO.

The results showed that models including drift rate modulations provided a better fit than models based only on starting point changes (Figure 4). This suggests that previous decisions do not only create an initial response bias before evidence processing begins, but also modulate how new sensory information is accumulated during the decision process.

From a computational perspective, this pattern is consistent with confirmation bias mechanisms in which earlier decisions bias the accumulation of later sensory evidence toward the previously selected interpretation.

6.2.4 Summary of computational results

The HSSM analyses complemented the behavioral findings and helped explain the computational processes underlying the observed bias. While the behavioral analyses showed a strong impact of prior decisions on later responses, the computational modeling helped explain how these effects emerged at the level of latent decision processes.

The simulations and parameter recovery analyses showed reliable recovery of the main model parameters, supporting the validity of the modeling approach. In addition, model comparison analyses suggested that the observed biases were better explained by drift rate modulations than by starting point changes alone. Overall, these results support the idea that perceptual confirmation bias reflects an influence on how new sensory evidence is accumulated during decision-making rather than only a simple tendency to repeat previous responses.

Finally, the computational framework developed in this stage provides a useful basis for future machine learning analyses and for the later study of EEG correlates associated with perceptual decision-making.

7. Discussion

7.1 Confirmation bias and evidence accumulation

The results obtained in this study provide strong evidence that previous decisions play an important role in how new sensory information is integrated during repeated perceptual decision-making tasks. Both the behavioral analyses and the computational models showed patterns consistent with confirmation bias mechanisms, where previous decisions bias the integration of later sensory evidence.

Behavioral results showed a strong tendency to maintain previous responses across successive repetitions of the stimulus, together with differential patterns in sensory evidence processing. These effects did not seem to depend

only on the exact repetition of the visual stimulus, suggesting that prior decisions seem to bias how participants interpret new perceptual information.

In addition, the HSSM analyses showed that models based on drift rate modulations provided a better fit to the data than models based only on changes in starting point. This suggests that the observed bias mainly emerges during the dynamic process of evidence accumulation rather than only as an initial predisposition before sensory processing begins.

This interpretation is consistent with recent studies proposing that perceptual confirmation bias may emerge from mechanisms that bias later evidence accumulation toward previous choices.

7.2 Metacognition, confidence and decisional stability

The results also suggest that metacognitive processes play an important role in the stability and revision of perceptual decisions. In particular, the relationship observed between subjective confidence and the probability of maintaining previous decisions indicates that internal confidence signals may alter how new sensory evidence is integrated after an initial choice.

Participants were more likely to maintain their responses when previous confidence was high, whereas changes of mind were associated with considerably lower confidence levels. This pattern is consistent with recent models suggesting that confidence may act as an internal signal related to decisional stability and flexibility.

Although global metacognitive efficiency measures did not show strong associations with the general tendency to repeat decisions, this result may partly be explained by the high response stability observed in most participants, which reduced the variability needed to detect clearer individual differences.

In addition, the analyses showed that changing a previous decision did not always lead to improvements in perceptual accuracy.

7.3 Limitations

Despite the consistency of the results obtained so far, the present work still has several limitations. Although the behavioral and computational analyses provide useful insights into perceptual confirmation bias, they do not directly identify the underlying neural processes, since no neural recordings have been analyzed at this stage of the project.

In addition, although HSSM models are useful for studying decision-making mechanisms, they represent simplified versions of perceptual processes. For this reason, the results obtained from these models should be interpreted cautiously and not as a direct description of how the brain exactly processes sensory information.

8. Current project status and future work

At the current stage of the project, the preprocessing and initial behavioral analyses of the data have been completed, together with the implementation of mixed-models and hierarchical evidence accumulation models (HSSM). The analyses carried out so far have identified consistent patterns of perceptual confirmation bias and provided an initial computational interpretation of the mechanisms involved in sensory evidence integration.

In addition, simulations and parameter recovery analyses have been performed to validate the computational models and compare different bias mechanisms related to starting point and drift rate modulations.

The next stages of the project will focus on developing machine learning approaches to predict participants' decisions from behavioral variables and computational model parameters. Later, EEG analyses will be incorporated to investigate the neural correlates associated with evidence accumulation and perceptual confirmation bias.

Finally, the last stages of the project will focus on integrating the different results, combining behavioral and neural analyses.

9. Bibliography

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Appendix. Computational Analyses (HSSM)

A1. Simulations and parameter recovery analyses

Before applying the HSSM models to the real behavioral data, simulations and parameter recovery analyses were performed to validate the computational pipeline and verify that the models could reliably recover the parameters used to generate the data.

Different versions of the DDM were implemented representing two possible confirmation bias mechanisms. In the first model, previous responses modified the starting point of the decision process, creating an initial bias before processing new sensory information. In the second model, previous decisions altered the drift rate, directly biasing the way new sensory evidence was integrated during decision-making tasks.

Using these models, simulated datasets were generated including perceptual decisions and response times similar to those observed in the real participants.

The simulations showed clear differences between both mechanisms.

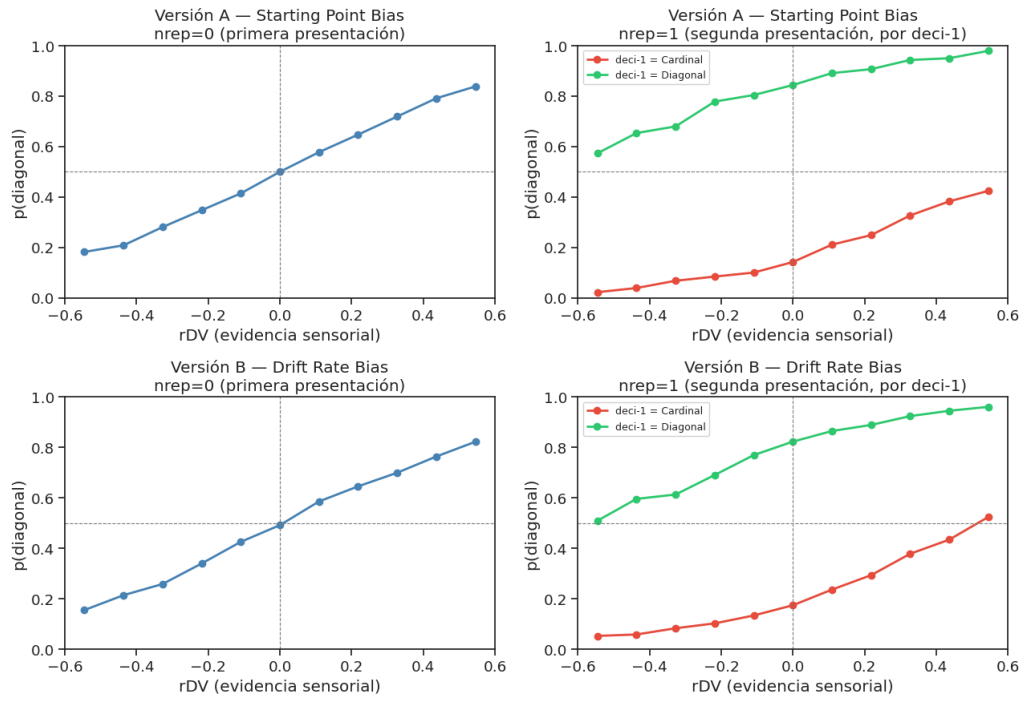


Figure A1. Simulated psychometric effects generated by starting point bias and drift-rate bias mechanisms within the Drift Diffusion Model framework. Drift-rate modulation produced stronger changes in perceptual sensitivity across repetitions.

Models based on starting point bias mainly affected response times, while models including drift-rate bias produced stronger effects on perceptual behavior and psychometric curves.

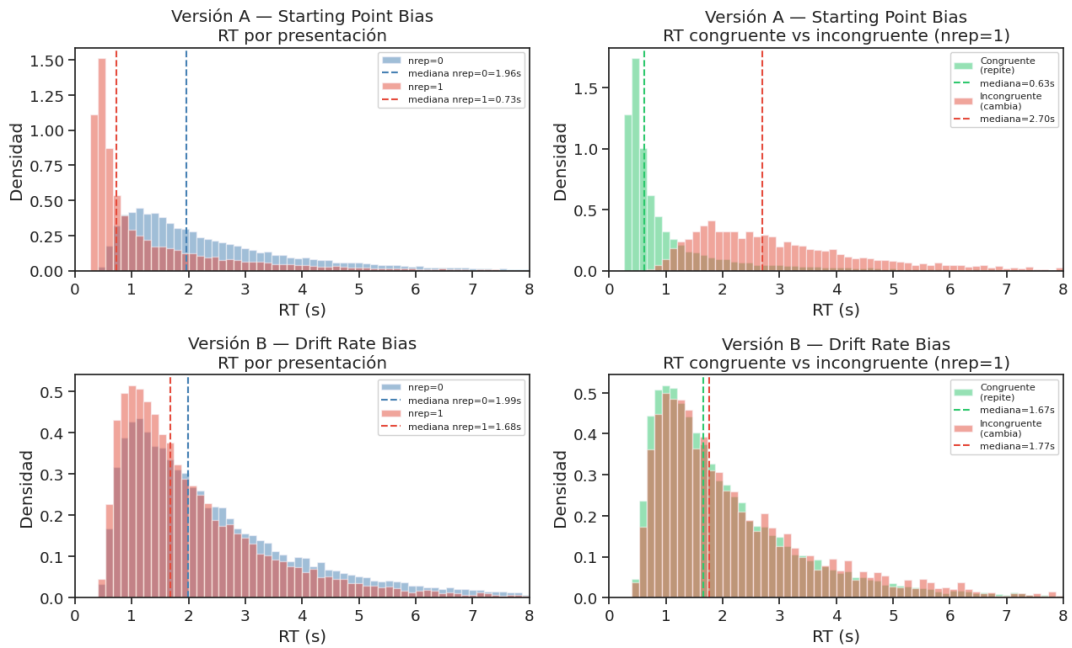


Figure A2. Simulated response time distributions for starting point bias and drift-rate bias models. Starting point bias produced stronger effects on response times, particularly between congruent and incongruent decisions.

Afterwards, the HSSM models were fitted again to the simulated datasets in order to evaluate whether the original parameters could be successfully recovered. The analyses showed good recovery of the main model parameters, indicating that the inference procedure was stable and reliable.

Overall, this stage allowed the computational pipeline to be validated before fitting the models to the real behavioral data.

A2. HSSM model comparison

After validating the models through simulations, the HSSM framework was applied to the real behavioral data obtained during the perceptual task. The main goal of this analysis was to investigate which computational mechanism better explained the behavioral biases observed in participants. To do so, models where previous decisions influenced different parameters of the evidence accumulation process were compared.

On the one hand, models in which previous decisions modified the starting point of the decision process were evaluated. On the other hand, models where previous decisions altered the drift rate were also tested, suggesting that previous decisions bias the processing of new sensory information.

The models were fitted using hierarchical Bayesian inference and included variables related to sensory evidence, previous decisions and subjective confidence.

Model comparison was performed using predictive metrics such as WAIC and LOO. The results showed that models including drift-rate modulations explained participants' behavior better than models based only on starting point changes.

Overall, these findings suggest that perceptual confirmation bias is not only related to an initial tendency to repeat previous decisions, but also to an active modification in the way new sensory information is processed during decision-making.